

AI FITTRACK

PHASE 2 – REQUIREMENT ANALYSIS

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1. PHASE OVERVIEW

Requirement analysis defines what the assigned AI FitTrack system is required to provide, the environment in which it operates, and the boundaries of the verified implementation. This phase translates the project case study and supplied implementation into clear functional and non-functional requirements.

The analysis is intentionally limited to the assigned FitTrack backend. It does not claim broader frontend, administrator, notification, or additional fitness-database functionality that is not present in the supplied implementation.

2. PROJECT OVERVIEW

AI FitTrack is a secure and modular backend REST API developed for workout tracking and AI-assisted fitness guidance. The backend uses Node.js and Express.js for the API layer, MongoDB with Mongoose for persistence, JWT for authentication, bcryptjs for password protection, and Google Gemini for personalized workout recommendations and fitness insights.

The supplied implementation is a backend API. API consumers such as Thunder Client or Postman communicate with the backend through HTTP requests and receive JSON responses. The verified database design contains two primary models: User and Workout.

3. PROBLEM STATEMENT

Fitness users may find it difficult to maintain a centralized workout history, search previous records, and determine whether their exercise activity aligns with their goals. Manual tracking can make historical records difficult to organize and does not inherently provide personalized guidance.

The system therefore requires authenticated access, structured workout records, user-level ownership checks, searchable workout information, and AI-assisted operations. The assigned FitTrack case study identifies workout management, search, AI recommendation, and AI fitness insights as core requirements.

4. FUNCTIONAL REQUIREMENTS

Requirement	Description
FR-01 – User Registration	The system shall allow a user to register using the authentication API.
FR-02 – User Login	The system shall authenticate registered users and provide a JWT token for protected API access.
FR-03 – Profile Access	The system shall allow an authenticated user to retrieve the user's profile.
FR-04 – Workout Creation	The system shall allow an authenticated user to create a workout record.
FR-05 – Workout Retrieval	The system shall allow an authenticated user to retrieve workout records.

Requirement	Description
FR-06 – Workout Search	The system shall support workout search by name, category, and date.
FR-07 – Workout Retrieval by ID	The system shall allow retrieval of a specific workout record using its identifier, subject to ownership checks.
FR-08 – Workout Update	The system shall allow an authenticated user to update a workout record.
FR-09 – Workout Deletion	The system shall allow an authenticated user to delete a workout record.
FR-10 – AI Workout Recommendation	The system shall accept age, fitness goal, and experience level and request a personalized workout recommendation through Google Gemini.
FR-11 – AI Fitness Insights	The system shall accept total workouts, average duration, and total calories burned and request AI-generated fitness insights.
FR-12 – JSON API Response	The system shall expose backend functions through RESTful HTTP endpoints and structured JSON responses.
FR-13 – Protected Operations	Protected workout and AI operations shall require valid JWT authentication.

5. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement
Security	Passwords shall be protected using bcryptjs hashing and protected API operations shall use JWT-based authentication.
Data Integrity	Workout records shall be associated with the authenticated user and validated through the Mongoose model.
Maintainability	The backend shall separate routes, controllers, middleware, models, configuration, and AI service responsibilities.
Interoperability	The system shall expose REST endpoints that can be consumed by API clients such as Thunder Client or Postman.
Performance	The API should process normal CRUD, search, and AI requests without unnecessary application-layer complexity.
Reliability	The application shall provide centralized error handling and clear API responses for failed requests.
Configurability	Environment-specific values such as port, MongoDB connection, JWT secret, and Gemini API key shall be supplied through environment variables.
Scalability	The modular backend structure shall allow additional clients or modules to be integrated without replacing the core API structure.

6. USER / ACTOR REQUIREMENTS

The verified FitTrack implementation identifies one primary application role: the authenticated user. The user interacts with the backend through an API consumer.

Actor	Required Interaction
Authenticated User	Register, log in, access profile, manage personal workout records, search workouts, request AI recommendations, and request fitness insights.
API Consumer	Send HTTP requests to the FitTrack REST API and display or inspect JSON responses. Thunder Client and Postman are supported API-testing clients.

7. HARDWARE AND SOFTWARE REQUIREMENTS

Type	Requirement
Operating System	Windows 10/11, macOS, or Linux
Runtime	Node.js and npm
Database	MongoDB Atlas or an available MongoDB connection
Database ODM	Mongoose
Code Editor	Visual Studio Code or equivalent
API Testing	Thunder Client or Postman
AI Service	Google Gemini API access and API key
Network	Internet connectivity for MongoDB Atlas and Google Gemini API access
Recommended Hardware	Intel Core i5 / AMD Ryzen 5 class processor, 8 GB RAM minimum, 16 GB recommended, and sufficient storage for the project and dependencies

8. TECHNOLOGY STACK

Layer / Tool	Purpose
Node.js	Backend runtime
Express.js	HTTP server, middleware, and routing
MongoDB Atlas	Persistent NoSQL data storage
Mongoose	Schemas, validation, and MongoDB interaction
JWT	Protected API authentication
bcryptjs	Password hashing and password comparison
Google Gemini / @google/genai	Workout recommendations and fitness insights
dotenv	Environment-variable loading
cors	Cross-origin request support
morgan	HTTP request logging
Thunder Client / Postman	API testing and verification
Visual Studio Code	Development environment

9. SYSTEM SCOPE

In Scope	Out of Current Scope / Future Extension
Node.js and Express.js backend	Separate React frontend
MongoDB and Mongoose	Administrator dashboard
User registration, login and profile	Advanced role-based administration
JWT authentication and bcryptjs password hashing	Notifications module
Workout CRUD operations	Additional fitness collections not present in the implementation
Workout search by name, category and date	Progress/goal/history collections beyond the supplied User and Workout models
Gemini workout recommendation	Production deployment and monitoring
Gemini fitness insights	Unverified third-party integrations
REST API and modular backend structure	Features not represented in the assigned project scope

Scope boundary: a broader generic system-requirements document contains additional frontend, database, and administrative features. Those broader items are not claimed here as implemented functionality unless they are part of the assigned FitTrack implementation.

10. API REQUIREMENT SUMMARY

Module	Method	Endpoint	Purpose
Authentication	POST	/api/auth/register	Register a user
Authentication	POST	/api/auth/login	Authenticate a user
Authentication	GET	/api/auth/profile	Retrieve authenticated profile
Workout	POST	/api/workouts	Create workout
Workout	GET	/api/workouts	Retrieve workouts
Workout	GET	/api/workouts/search	Search by name, category, or date
Workout	GET	/api/workouts/:id	Retrieve a workout by ID
Workout	PUT	/api/workouts/:id	Update a workout
Workout	DELETE	/api/workouts/:id	Delete a workout
AI	POST	/api/ai/workout-recommendation	Generate workout recommendation
AI	POST	/api/ai/fitness-insights	Generate fitness insights

11. REQUIREMENT TRACEABILITY

Requirement Area	Implementation Evidence / Component
Authentication	authRoutes.js, authController.js, auth.js middleware, User.js
Workout Management	workoutRoutes.js, workoutController.js, Workout.js
Workout Search	workoutController.js and GET /api/workouts/search
Database	config/db.js, User.js, Workout.js, MongoDB Atlas
AI Recommendation	aiRoutes.js, aiController.js, geminiService.js
AI Insights	aiRoutes.js, aiController.js, geminiService.js
Application Structure	src/app.js, src/server.js, routes, controllers, middleware, models, services, config
Error Handling	middleware/errorHandler.js and Express application configuration
API Verification	FitTrack Postman collection and Thunder Client/Postman testing workflow

12. SYSTEM LIMITATIONS IDENTIFIED DURING REQUIREMENT ANALYSIS

The supplied project is a backend implementation rather than a complete end-user frontend application. The verified database design contains User and Workout models only. AI outputs depend on access to the configured Google Gemini service. Production deployment, monitoring, administrator functionality, and broader fitness modules are outside the verified current scope.

These limitations are recorded as scope boundaries rather than defects in the assigned project. Future enhancements may extend the system after the current academic implementation is completed.

13. REQUIREMENT ANALYSIS CONCLUSION

The requirement analysis establishes a clear boundary for AI FitTrack: a modular Node.js and Express.js REST backend with MongoDB/Mongoose persistence, JWT authentication, bcryptjs password protection, workout CRUD and search operations, and Gemini-powered recommendation and fitness-insight endpoints.